

Appendices

Pricing Intelligence & Revenue Optimisation

Supplementary material supporting the Executive Report

This document contains three appendices:

- Appendix A: Analytical Framework and Data Coverage
 - Appendix B: Complete Opportunity Register (15 opportunities)
 - Appendix C: Simulation Methodology and Assumptions (12 modelled scenarios)
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Appendix A: Analytical Framework and Data Coverage

Data Source and Coverage

Dataset: Indian E-Commerce Pricing, Revenue and Growth Dataset

Period: 36 months (January 2023 to December 2025)

States: 14 states across five zones (South, West, North, East, Central)

Categories: 8 product categories: Electronics, Fashion, Footwear, Grocery Essentials, Home & Living, Premium Lifestyle, Sports & Fitness, Beauty & Personal Care

Brand tiers: Mass and Premium

Segments analysed: 224 (state x category x brand tier combinations)

Discount response segments: 160 (state x category x brand tier x period combinations)

Five Analytical Dimensions

1. Business Performance Assessment Goal: Understand overall revenue and volume trends across time and geography. Evaluated monthly revenue growth, unit sales trends, geographic distribution, and state-level share dynamics to assess whether growth was stable, slowing, or increasingly concentrated in a limited number of high-performing markets.

2. Pricing and Discount Evaluation Goal: Examine how different discounting strategies influence revenue generation and margin quality. Discount performance was assessed across multiple discount bands (from below 10% to above 70%) to identify which pricing conditions generated the strongest commercial outcomes. The analysis incorporated discount cost ratios (DCR) and realisation ratios to distinguish commercially efficient discounting from value-destructive promotional activity.

3. Demand Response Analysis Goal: Understand how customers respond to pricing changes under different market and operational conditions. Customer demand behaviour was evaluated across age groups, product categories, geographic regions, promotional environments, competition intensity levels, and inventory pressure conditions. This dimension identified where pricing sensitivity was strongest and where discounting was failing to generate meaningful volume response.

4. Revenue Quality and Sustainability Assessment Goal: Evaluate the sustainability of revenue performance beyond top-line growth metrics. Revenue quality indicators were used to distinguish between stable growth and discount-dependent or event-driven performance. This dimension introduced a strategic risk perspective by identifying segments where pricing sustainability appeared to be weakening despite strong headline sales.

5. Strategic Risk and Opportunity Identification Goal: Consolidate findings across all four prior dimensions to identify high-performing segments, high-risk revenue pockets, and areas requiring strategic pricing review. Segments were classified into risk tiers (Urgent Review, Monitor, High Risk) and evaluated for intervention priority based on revenue significance, discount dependency, demand volatility, and competitive exposure.

SQL Views Used in Analysis

The following SQL views were developed to support each analytical dimension. Each view aggregates and structures raw transaction data for a specific analytical purpose.

SQL View	Analytical Dimension	Purpose
V_monthly_revenue_summary	Business Performance	Monthly revenue, unit volume, and growth trends across the 36-month study period.
V_state_growth	Business Performance	State-level revenue growth and share distribution across all 14 tracked states.
V_revenue_concentration_geo	Business Performance	Geographic revenue concentration analysis, including coverage curves and dependency risk.
V_monthly_discount_band	Pricing and Discount Evaluation	Monthly unit and revenue distribution across discount bands from below 10% to 70% and above.
V_discount_band_performance	Pricing and Discount Evaluation	Revenue per unit, realisation ratio, and discount cost ratio by discount band and brand type.
V_pricing_intelligence	Pricing and Discount Evaluation	Brand-level pricing intelligence including optimal discount band identification and DCR benchmarks.
V_discount_response_quality	Demand Response	Positive discount response rates across 160 segments, measuring how reliably discount increases generate unit increases.

SQL View	Analytical Dimension	Purpose
V_price_sensitivity_by_segment	Demand Response	Elasticity proxies and price sensitivity classification by age group, brand type, category, and state.
V_competition_discount_performance	Demand Response	Discount depth and realisation outcomes segmented by competition intensity level (Low, Medium, High).
V_inventory_pricing_pressure	Demand Response	Discount behaviour and revenue outcomes under high versus low inventory pressure conditions.
V_event_uplift_by_category	Revenue Quality	Festival period versus normal period revenue and unit comparisons by category and brand type.
V_risk_adjusted_performance	Strategic Risk	Segment-level risk scoring across 224 combinations of state, category, and brand tier.
V_risk_adjusted_summary	Strategic Risk	Portfolio-level aggregation of risk tiers: Protect and Scale, Monitor, Urgent Review, and High risk.
V_revenue_quality_risk_hotspots	Strategic Risk	Identification of segments with the highest concentration of discount dependency, revenue volatility, and structural fragility.

Note: All views are built on the same underlying transaction dataset. Cross-view consistency was validated prior to analysis.

Appendix B: Complete Opportunity Register

The 15 opportunities identified through the analysis are listed in full below, grouped by theme. The main report consolidates these into five priority actions and a further investigation list. This appendix provides the complete register for reference.

Implementation tiers are defined as follows:

- Immediate: Policy or governance change with zero operational cost. Can be implemented in Months 1 to 2 without systems changes or data infrastructure work.

- Short-term: Operationally viable within the current planning cycle. Requires baseline setting, validation, or moderate process change before execution.
- Investigation: Analytically supported but requires additional data, audit, or sub-segment targeting before a go/no-go decision can be made.
- Deferred: Simulation results were negative at the portfolio level. Requires live elasticity data from earlier implementations before recalibration and retry.

#	Opportunity	Theme	Simulation	Tier
Discount Governance				
1	Introduce a hard discount ceiling at 40% for Mass brand segments, requiring escalation approval before any transaction is routed into the 51-70% bands.	Discount Governance	OPP_01/02/03	Deferred
2	Reduce average discount depth in Electronics (Mass) from 45.7% to the 31-40% optimal band to recover realisation without meaningful volume risk.	Discount Governance	OPP_01/02/03	Short-term
6	Ringfence Premium brand discount depth at 30% or below across all demographics and categories to protect the margin profile that differentiates it from Mass.	Discount Governance	OPP_06	Immediate
1 1	Develop a category-level discount response policy distinguishing price-elastic categories (Grocery Essentials, Beauty & Personal Care) from price-inelastic ones, applying discounts selectively rather than uniformly.	Discount Governance	OPP_09 / OPP_11	Investigation
1 5	Establish realisation ratio and discount cost ratio targets at the segment level as performance metrics alongside revenue, so that margin quality is tracked with the same rigour as top-line growth.	Discount Governance	Not modelled	Short-term
Festival and Events Strategy				
7	Redirect festival promotional spend toward normal-period demand stimulation in categories where organic uplift already occurs, particularly Sports & Fitness, Home & Living, and Electronics.	Festival Strategy	OPP_10	Short-term

#	Opportunity	Theme	Simulation	Tier
13	Treat the triple-risk combination of festival period, high competition, and deep discounting as a controlled spend decision with a defined budget ceiling rather than a default commercial response.	Festival Strategy	OPP_08	Investigation
14	Review the commercial rationale for discounting Premium Lifestyle and Electronics during festival periods, given both categories consistently underperform their normal-period baseline during events.	Festival Strategy	OPP_04	Short-term
Customer Segmentation				
5	Gradually introduce full-price exposure in Elderly and Middle-aged customer segments beginning in low-elasticity categories, implementing in three steps to test and recover margin from promotionally conditioned groups.	Customer Segmentation	OPP_05	Immediate
3	Build a rules-based promotional calendar for Mass Fashion and Mass Premium Lifestyle to prevent discretionary markdown decisions in the two highest price-sensitivity categories.	Promotional Planning	Not modelled	Investigation
Inventory and Markdown Management				
8	Adopt the 31-40% band as the default ceiling for all inventory-driven markdowns, with a structured authorisation process required before deeper clearance cuts are approved.	Inventory Management	OPP_07	Investigation
9	Benchmark the markdown governance practices of Home & Living across all other categories, given its consistent outperformance on realisation under both inventory and competitive pressure.	Markdown Governance	Not modelled	Investigation
Geographic and Risk Intelligence				
4	Conduct a dedicated pricing audit in Gujarat to identify and address the state-specific dynamic driving the highest discount cost ratio in the portfolio (DCR 1.547).	Geographic Intelligence	Not modelled	Investigation
10	Investigate the six Medium-risk Mass brand segments to identify what differentiates their stability from structurally similar Urgent Review counterparts, with a view to replicating those conditions elsewhere.	Risk Management	Not modelled	Investigation
Data Infrastructure				

#	Opportunity	Theme	Simulation	Tier
1 2	Prioritise product-level data collection to resolve the unit-revenue divergence question, as the root cause, likely a product mix shift, cannot be confirmed or acted on without granular category data.	Data Infrastructure	<i>Not modelled</i>	Investigation

Note: Numbering follows the original sequence from Section 6 of the Executive Report. Items 1-15 correspond directly to that section.

Simulation-identified opportunities not in the original 15:

Two additional opportunities were surfaced through simulation modelling and became Final Recommendations in the main report: OPP_09 (Price-Inelastic Segments to Optimal Band) and OPP_12 (Lower-Tier State Demand Activation). OPP_09 produced the strongest single result in the simulation (+7.6%, +₹166M). OPP_12 is the only opportunity with zero unit loss risk and is treated as a pure volume upside play.

Appendix C: Simulation Methodology and Assumptions

Overview

The simulation used 36 months of historical transaction data to estimate the revenue and realisation impact of 12 pricing scenarios. Each scenario models a specific change to discount depth, segment targeting, or demand activation, and calculates the resulting revenue delta against the historical baseline.

The general modelling logic across all scenarios follows three steps:

- Step 1: Identify the affected segment rows within the historical dataset based on the scenario definition (brand type, discount band, age group, state, or event period).
- Step 2: Apply the proposed change (discount reduction, demand growth rate, or ceiling) to those rows.
- Step 3: Calculate the new revenue using an elasticity-adjusted unit estimate and the revised price, then compare against the historical revenue for those rows to derive the percentage delta.

Elasticity proxies were derived from observed discount-demand relationships within the dataset rather than assumed from external benchmarks. Where a specific elasticity value is noted below, it reflects the best-fit estimate from the segment data. Where elasticity is described qualitatively, the dataset did not yield a precise numerical estimate for that specific scenario scope.

Scenario Results and Assumptions

Reference	Scope	Elasticity	Modelling Basis	Outcome and Key Limitation
Positive Revenue Impact				
OPP_09	Home & Living, Footwear, Premium Lifestyle	0.2 (low)	Discount depth reduced from above 40% to the data-derived optimal of 30.76%. Revenue and unit change calculated using observed elasticity proxy for each segment.	+7.6% (+₹166M). Unit loss: -11.6%. Strongest single result in the simulation.
OPP_05	Elderly and Middle-Aged segments (all categories)	Low (not specified exactly)	Three-step migration: current discounts reduced to 30%, then 25%, then 20% caps. Each step modelled independently. Revenue delta calculated against current segment baseline.	+3.7% (Step 1), +4.6% (Step 2), +5.5% (+₹119M) (Step 3). Unit loss: -11.6% at Step 3.

Reference	Scope	Elasticity	Modelling Basis	Outcome and Key Limitation
OPP_12	Uttar Pradesh, West Bengal, Odisha	N/A	Demand growth modelling only: no discount change applied. Revenue gap between lower-tier and top-performing states closed at 10%, 20%, and 30% increments.	+1.7%, +3.4%, +5.1% (+₹111M) respectively. Zero unit loss risk.
OPP_06	All Premium brand segments	Low (DCR 0.437 as proxy)	Hard ceiling applied at 30% across all Premium brand promotional activity. Revenue recalculated using Premium segment realisation ratios (0.69-0.71) and observed unit response.	+1.2% (+₹26M). Unit loss: -5.5%.
OPP_04	Electronics (Mass) and Premium Lifestyle during festival periods	Segment-specific	Festival-period discount rates returned to their normal-period weighted averages. Revenue delta measured against the festival baseline for each category.	+0.68% (Electronics) and +0.67% (Premium Lifestyle), +₹15M combined. Unit loss under -8%.
OPP_07	All high-inventory-pressure segments	Segment-specific	Inventory-driven markdowns capped at 40%. Units that previously cleared at 51-70% are assumed to clear at 40% or remain as uncleared stock.	+0.44% (+₹9.5M). Unit cost: -24.7%.
OPP_10	All categories (portfolio-wide)	Portfolio average	Discount reductions applied simultaneously to all 30,600 rows in the dataset. Festival-period discounts reduced to normal-period levels across all categories.	+14.9% (+₹324M). Treated as directional upper bound only.
Negative or Not Viable				
OPP_08	Festival periods with high competition and deep discounting (₹137M in revenue)	High (competitive context)	Deep discounts removed from the triple-risk combination of festival period, high competition intensity, and 51-70% discount depth. Revenue delta calculated against this subset baseline.	-0.13%. Discounts appear to be doing real competitive work in this combination.
OPP_11	Grocery Essentials (Mass)	0.9 (high)	Three-step discount reduction modelled for Grocery Essentials. Volume response estimated using the 0.9	-0.26% to -0.31% across all three steps. Volume loss outpaces price recovery at every level.

Reference	Scope	Elasticity	Modelling Basis	Outcome and Key Limitation
			elasticity proxy derived from segment data.	
OPP_01/02/03	All Mass brand segments (progressive discount caps from 50% down to 31%)	0.8 (moderately high)	Progressive caps applied to Mass brand discount depth in three steps. Each step modelled independently against the Mass brand revenue baseline.	All steps negative, worsening at each step. A single-step policy cap does not recover revenue at 0.8 elasticity.

General Limitations Applicable to All Scenarios

- The simulation uses historical data as a static baseline. It does not model competitor responses to pricing changes, which may amplify or dampen the estimated impact in live conditions.
- Elasticity estimates are derived from observed historical behaviour and may not hold if the business moves outside the observed discount range for a given segment.
- Scenarios are modelled independently. Simultaneous implementation of multiple opportunities could produce interaction effects (positive or negative) not captured in the individual results.
- The simulation does not account for implementation costs such as system changes, staff training, campaign redesign, or the cost of market activation activity in lower-tier states.
- Revenue deltas are expressed as portfolio-level percentages. Individual segment outcomes may vary significantly from the portfolio average, particularly in scenarios where only a subset of segments is affected.
- Downstream effects (such as inventory accumulation from reduced clearance discounting in OPP_07) are not modelled. Results should be interpreted as first-order revenue impacts only.

Recommended Validation Approach

The simulation provides directional confidence, not certainty. The following validation steps are recommended before committing to full-scale implementation of any scenario:

- Phase 1 and 2 policy changes (OPP_06, OPP_09): Compare actual realisation ratios and unit volumes at Month 8 against simulation projections. Revise elasticity assumptions if real-world response diverges by more than 20%.
- OPP_05 (age segment migration): Measure unit and revenue response after each step before advancing. Do not proceed to Step 2 if Step 1 unit loss exceeds -10%.
- OPP_12 (lower-tier states): Launch wave one in a single state to establish an activation cost-to-revenue ratio before scaling. Adjust the portfolio impact estimate accordingly.
- OPP_01/02/03 (Mass brand caps): Re-run the simulation in Month 19-24 using live elasticity data captured from OPP_05 and OPP_09 implementations. The observed price-response behaviour from those programmes will provide a more accurate elasticity input for Mass brand sub-segment modelling.